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Course: CyberOps

Lab: Linux Servers

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Screenshot 1 – Displaying Running Processes

The `sudo ps -elf` command was used to display all running processes on the Linux system.

Q:1

Why was it necessary to run **ps** as root (prefacing the command with **sudo**)?

It was necessary to run **ps** as root because some processes are owned by other users or the root user. Using **sudo** allows us to view complete

Screenshot 2 – Displaying Running Processes

```
(elghorab@kali)-[~]
$ sudo ps -ejH
  PID    PGID    SID TTY          TIME CMD
    2         0      0 ?           00:00:00 kthreadd
    3         0      0 ?           00:00:00 pool_workqueue_release
    4         0      0 ?           00:00:00 kworker/R-kvfree_rcu_reclaim
    5         0      0 ?           00:00:00 kworker/R-rcu_gp
    6         0      0 ?           00:00:00 kworker/R-sync_wq
    7         0      0 ?           00:00:00 kworker/R-slub_flushwq
    8         0      0 ?           00:00:00 kworker/R-netns
   11         0      0 ?           00:00:00 kworker/0:0H-events_highpri
   12         0      0 ?           00:00:00 kworker/u512:0-ipv6_addrconf
   13         0      0 ?           00:00:00 kworker/R-mm_percpu_wq
   14         0      0 ?           00:00:00 rcu_tasks_kthread
   15         0      0 ?           00:00:00 rcu_tasks_rude_kthread
   16         0      0 ?           00:00:00 rcu_tasks_trace_kthread
   17         0      0 ?           00:00:00 ksoftirqd/0
   18         0      0 ?           00:00:02 rcu_preempt
```

Note: Nginx was not installed on the system initially, so I installed it manually before continuing with the lab steps.

```
(elghorab@kali)-[~]
$ sudo /user/sbin/nginx
sudo: /user/sbin/nginx: command not found

(elghorab@kali)-[~]
$ pwd
/home/elghorab

(elghorab@kali)-[~]
$ sudo apt install nginx -y
Upgrading:
  nginx nginx-common

Summary:
  Upgrading: 2, Installing: 0, Removing: 0, Not Upgrading: 2282
  Download size: 756 kB
  Space needed: 99.3 kB / 3,480 MB available

Get:1 http://kali.download/kali kali-rolling/main amd64 nginx amd64 1.30.1-7 [647 kB]
Get:2 http://kali.mirror.garr.it/kali kali-rolling/main amd64 nginx-common all 1.30.1-7 [109 kB]
Fetched 756 kB in 3s (282 kB/s)
Preconfiguring packages ...
(Reading database ... 412542 files and directories currently installed.)
Preparing to unpack .../nginx_1.30.1-7_amd64.deb ...
Unpacking nginx (1.30.1-7) over (1.26.3-2) ...
Preparing to unpack .../nginx-common_1.30.1-7_all.deb ...
Unpacking nginx-common (1.30.1-7) over (1.26.3-2) ...
Setting up nginx-common (1.30.1-7) ...
Installing new version of config file /etc/nginx/fastcgi_params ...
Installing new version of config file /etc/nginx/proxy_params ...
Installing new version of config file /etc/nginx/scgi_params ...
Installing new version of config file /etc/nginx/uwsgi_params ...
Setting up nginx (1.30.1-7) ...
Processing triggers for kali-menu (2025.2.7) ...
Processing triggers for man-db (2.13.1-1) ...

(elghorab@kali)-[~]
$ sudo /user/sbin/nginx
sudo: /user/sbin/nginx: command not found

(elghorab@kali)-[~]
$ nginx -v
nginx version: nginx/1.30.1
```

First, I installed **Nginx** using the apt package manager. After the installation was completed successfully, I checked the installed Nginx version using the `nginx -v` command. The installed version was **Nginx 1.30.1**.

Then, I started the Nginx web server and displayed its information. The commands were executed successfully, and the output is shown in the screenshot below.

```
104106 104106 104105 pts/4    00:00:00 ps
103868 103868 103868 ?        00:00:00 nginx
103869 103868 103868 ?        00:00:00 nginx
103870 103868 103868 ?        00:00:00 nginx
```

Q:2:- How is the process hierarchy represented by ps

The process hierarchy is represented by showing parent and child processes. The child process is indented under its parent process.

```
(elghorab@kali)-[~]
$ netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
udp        0      0 192.168.88.131:bootpc   192.168.88.254:bootps   ESTABLISHED
Active UNIX domain sockets (w/o servers)
Proto RefCnt Flags   Type       State         I-Node  Path
unix   3      [ ]      STREAM    CONNECTED    38627     @/tmp/.X11-unix/X0
unix   3      [ ]      STREAM    CONNECTED    14128     /run/user/1000/at-spi/bus_0
unix   3      [ ]      STREAM    CONNECTED    13396     /run/systemd/journal/stdout
unix   3      [ ]      STREAM    CONNECTED    12044     /run/user/1000/bus
unix   3      [ ]      DGRAM     CONNECTED    8752      /tmp/dbus-7p4PMxrU3w
unix   3      [ ]      STREAM    CONNECTED    38629     /tmp/dbus-7p4PMxrU3w
unix   3      [ ]      STREAM    CONNECTED    14118     /run/user/1000/bus
unix   3      [ ]      STREAM    CONNECTED    14604     /run/user/1000/bus
unix   3      [ ]      STREAM    CONNECTED    8930      /run/dbus/system_bus_socket
unix   3      [ ]      STREAM    CONNECTED    38640     @/tmp/.X11-unix/X0
unix   3      [ ]      STREAM    CONNECTED    13386     /run/systemd/journal/stdout
unix   3      [ ]      STREAM    CONNECTED    12499     /run/systemd/journal/stdout
unix   3      [ ]      STREAM    CONNECTED    14820     /run/dbus/system_bus_socket
unix   3      [ ]      STREAM    CONNECTED    14664     /run/dbus/system_bus_socket
unix   3      [ ]      STREAM    CONNECTED    12966     /run/dbus/system_bus_socket
unix   3      [ ]      STREAM    CONNECTED    14107     /run/dbus/system_bus_socket
unix   2      [ ]      STREAM    CONNECTED    9036     /run/dbus/system_bus_socket

(elghorab@kali)-[~]
$ sudo netstat -tunap
[sudo] password for elghorab:
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 192.168.88.131:8080     0.0.0.0:*               LISTEN      79367/python3
tcp        0      0 0.0.0.0:80             0.0.0.0:*               LISTEN      103868/nginx: maste
tcp6       0      0 :::80                  :::*                    LISTEN      103868/nginx: maste
udp        0      0 192.168.88.131:68      192.168.88.254:67      ESTABLISHED 737/NetworkManager
```

Q3:- What is the meaning of the -t, -u, -n, -a and -p options in netstat? (use man netstat to answer)

- t --> TCP connections
- u --> UDP connections
- n --> Show numerical addresses/ports
- a --> Show all connections and listening ports
- p --> Show PID/program name

Q4:- Is the order of the options important to netstat?

No. The order of these options is not important as long as they are provided as options to netstat

Based on the netstat output shown in item (d), what is the Layer 4 protocol, connection status, and PID of the process running on port 80?

Protocol: TCP
Status: LISTEN
PID: 103868

Q:4:-What kind of service is running on port 80 TCP?

HTTP web server.

```

(elghorab@kali) [~]
$ sudo ps -elf | grep 103868
[sudo] password for elghorab:
1 S root      103868      1  0  80  0 - 3794 sigsus 18:55 ?        00:00:00 nginx: master process /usr/sbin/nginx
5 S www-data 103869      103868 0  80  0 - 4323 do_epo 18:55 ?        00:00:00 nginx: worker process
5 S www-data 103870      103868 0  80  0 - 4323 do_epo 18:55 ?        00:00:00 nginx: worker process
0 S elghorab 125069     95821  0  80  0 - 1632 pipe_r 19:37 pts/3    00:00:00 grep --color=auto 103868

```

Q5:- How could it be concluded that the process PID is nginx?

The ps output shows the nginx process name and command associated with that PID, including nginx master process

Q6:-What does it mean that process 396 has process 395 as its parent?

It means that process 395 is the parent process that started process 396. This is common because servers such as nginx commonly use a master process that creates worker processes.

Q7:- Why is the last line showing grep 395?

Because the grep command itself contains the number 395, so the ps output also matches the grep search.

```

helloConnection closed by foreign host.

(elghorab@kali) [~]
$ telnet 127.0.0.1 80
Trying 127.0.0.1...
Connected to 127.0.0.1.
Escape character is '^]'.
hello
HTTP/1.1 400 Bad Request
Server: nginx
Date: Fri, 14 Aug 2026 17:21:27 GMT
Content-Type: text/html
Content-Length: 150
Connection: close

<html>
<head><title>400 Bad Request</title></head>
<body>
<center><h1>400 Bad Request</h1></center>
<hr><center>nginx</center>
</body>
</html>
Connection closed by foreign host.

```

Q8:- Why was the error sent as a web page

Because nginx is a web server and it responds to HTTP requests using HTTP responses formatted as web pages.

```
(elghorab@kali)-[~]  
$ telnet 127.0.0.1 22  
Trying 127.0.0.1 ...  
telnet: Unable to connect to remote host: Connection refused  
  
(elghorab@kali)-[~]
```

Telnet Test on Port 22:

I attempted to connect to the local host on TCP port 22 using Telnet. The connection was refused, indicating that no service was listening on port 22 at the time of the test.

```
(elghorab@kali)-[~]  
$ telnet 127.0.0.1 68  
Trying 127.0.0.1 ...  
telnet: Unable to connect to remote host: Connection refused
```

Telnet Test on Port 68

I attempted to connect to the local host on TCP port 68 using Telnet. The connection was refused, indicating that no service was listening on port 68 at the time of the test.

1. What are the advantages of using netstat?

Netstat can display active network connections, listening ports, protocols, local and remote addresses, connection states, and the PID/program associated with network services.

2. What are the advantages of using Telnet? Is it safe?

Telnet can be useful for testing TCP services and identifying the type of service running on a specific port. However, Telnet is not safe for remote administration because it does not encrypt session data and can expose